

A
Project Report On
GYM PRO – GYM MANAGEMENT SYSTEM

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By

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CERTIFICATE

(To be provided by the Project Guide on completion of the project)

Preface

This report contains all the outputs of a project undertaken to develop a working model of **GYM PRO – Gym Management System** and to study the existing working system. As part of this study, we analyzed the requirements of the working model and tried to improve it using the technological functionality that the current computer science provides.

This volume contains the documentation summary of the working system, the background study and analysis, the technical requirements and the working specification of the developed system.

We recommend the user to take the advantage of the study presented here and implement the system with their required improvisation. We hope this study will reduce the burden of the reanalysis of the development phase.

Acknowledgment

“It is not possible to prepare a project report without the assistance & encouragement of other people. This one is certainly no exception.”

On the very outset of this report, we would like to extend our sincere & heartfelt obligation towards all the personages who have helped us in this endeavor. Without their active guidance, help, cooperation & encouragement, we would not have made headway in the project. We are ineffably indebted to our HOD / Prof. Jitendra Kabariya for conscientious guidance and encouragement to accomplish this assignment.

We are extremely thankful and pay our gratitude to our project guide Prof. Ishita mem for her valuable guidance and support on completion of this project in its entirety.

We extend our gratitude to the Department Of Computer Science, Late P.S. Hirapara Mahila BCA College-Jetpur for giving us this opportunity.

Thanking You

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1. Project Profile

1.1 Definition

GYM PRO – Gym Management System is a web-based software application developed to manage and maintain the daily activities of a gym. It helps the gym administrator manage member information, membership details, batches, trainers, attendance, payments and other gym-related records in a systematic and efficient way.

The system reduces manual paperwork, saves time, improves data management, and makes it easier to store, update, and retrieve gym information whenever required. It provides three separate, password-protected panels for the Admin, the Trainers and the Members, in addition to a public informational website for visitors.

1.2 Scope

The **GYM PRO – Gym Management System** is designed to manage gym members, memberships, batches, trainers, attendance and payments in an organized way. It reduces manual paperwork and makes daily gym management easier for the owner and staff.

The system helps the administrator to add, update, delete and view records easily. It saves time, reduces errors, and provides quick access to important gym information such as who is due for renewal, which batch is running short of trainers, and how much revenue has been collected.

1.3 Objective

The main objective of the **GYM PRO – Gym Management System** is to develop a simple and efficient system for managing gym activities and records. It aims to reduce manual work and make gym management faster, easier, and more organized.

The major objectives are to:

- Maintain accurate **member and membership records**.
- Manage **batches, trainers, attendance and payment information**.
- Allow a member to join more than one batch while **automatically preventing any clash** between the timings of the selected batches.
- Provide easy **adding, updating, approving, deleting and viewing** of records.
- Reduce **paperwork and manual errors**.
- Save time and improve the overall **efficiency of gym management**.

1.4 Technical Description

Technical Aspect	Description
Front End	HTML5, CSS3, Bootstrap 5, Bootstrap Icons, JavaScript, Chart.js
Back End	PHP (procedural style, mysqli extension, prepared statements)
Database	MySQL

2. System Analysis and Specification

2.1 Study of Current System

Most small and medium gyms still manage their day-to-day operations manually. During the study of the existing system, the following observations were made:

- New member details, batch timings and fee amounts are written down in a membership register or note-book.
- Attendance of members is taken on a paper sign-in sheet kept at the reception desk.
- Trainers keep track of which members belong to which batch from memory or hand-written batch sheets pinned on a notice board.
- Renewal of a membership is tracked by checking a physical register or a diary maintained by the gym owner.
- Fee receipts are hand-written and a carbon copy or a simple cash-book is kept for accounts.

2.2 Weakness of Current System

The manual system currently followed by most gyms suffers from the following drawbacks:

- Registers can be lost, damaged, misplaced or duplicated, and there is no backup of the data.
- Checking whether a new admission's chosen time slot clashes with an existing batch is done manually and is error-prone; double-booking of a batch beyond its capacity is common.
- Retrieving a particular member's payment history or attendance record takes considerable time when done manually.
- There is no way for a member to check their own membership status, batch timing, attendance or dues without visiting or calling the gym.
- Sending renewal reminders or announcements to all members has to be done individually by phone or word of mouth, which is slow and unreliable.
- Trainers have no organized way to record attendance or maintain a workout plan for their batch.

2.3 Introduction of New System

GYM PRO – Gym Management System replaces the manual register with a centralized, web-based application. The new system introduces the following improvements:

- A public website lets any visitor view the gym's batches, trainers and gallery, and submit an online registration request; the Admin reviews and approves it, after which the member automatically receives login access.
- A member can join more than one batch at the same time, and the system automatically prevents the member from picking two batches whose timings overlap.
- Membership renewal is requested online by the member and approved by the Admin, with the expiry date extended automatically on approval.

- Trainers mark attendance for their own assigned batches from their own login, and can maintain a simple workout plan for the batch.
- Payments, equipment and announcements are all managed digitally through the Admin panel and can be viewed by the concerned member or trainer at any time.
- A dashboard gives the Admin an at-a-glance view of active members, pending requests and revenue collected.

2.4 Hardware and Software Requirement (Feasibility Study)

Software Requirement

Software	Specification
Front End	HTML5, CSS3, Bootstrap 5, JavaScript, Chart.js
Back End	PHP (compatible with PHP 5.4 as well as PHP 7/8)
Database	MySQL
Web Server	Apache (via XAMPP / WAMP / UwAmp)
Operating System	Windows 10 / 11 (or any OS that can run XAMPP/WAMP)
Browser	Google Chrome / Mozilla Firefox / Microsoft Edge (any modern browser)

Hardware Requirement

Hardware	Specification
Processor	Intel Pentium / Core i3 or above
RAM	4 GB minimum (8 GB recommended)
Hard Disk	100 MB free space for the project and the XAMPP/WAMP installation
Monitor	Standard colour monitor, 1024 x 768 resolution or higher
Input Devices	Keyboard, Mouse

3. System Requirement Specification

3.1 Proposed System and Advantages

Based on the study of the current manual system and its weaknesses, the following project model (modules) is proposed for GYM PRO – Gym Management System:

1) Public Website

Provides Home, About, Batches, Gallery and Contact pages so that any visitor can learn about the gym before joining.

2) Visitor Registration

A visitor can register online by filling in personal details and choosing one or more batches. The system groups batches by time-of-day and blocks any selection whose timings overlap, before the request is sent for approval.

3) Admin Module

The Admin can manage members, trainers, batches and equipment; approve or reject registration and renewal requests; record payments; post announcements; and view a dashboard of key statistics.

4) Trainer Module

A trainer can view the batches and members assigned to them, mark daily attendance, maintain a simple workout plan for the batch, and post announcements.

5) Member Module

A member can view their profile, batch(es), attendance and payment history; request renewal of an expiring membership; and view announcements posted by the Admin or trainer.

Advantages of the Proposed System

- Replaces scattered paper registers with a single, centralized, always-available database.
- Automatically prevents a member from being enrolled in two batches whose timings clash, which is not practically possible to check by hand.
- Cuts down the time needed to search for a member's record, payment history or attendance from minutes to seconds.
- Gives every member and trainer self-service, read-only access to their own information, reducing the number of enquiries handled at the reception desk.
- Keeps an accurate, tamper-resistant log of payments and approvals, since every action is tied to a logged-in Admin/Trainer account.
- Scales easily as the number of members, batches or trainers grows, unlike a paper register.

4. System Design

4.1 Use – Case Diagram

The system has four types of actors — Visitor, Member, Trainer and Admin. The use-case diagrams below show the functions available to each actor.

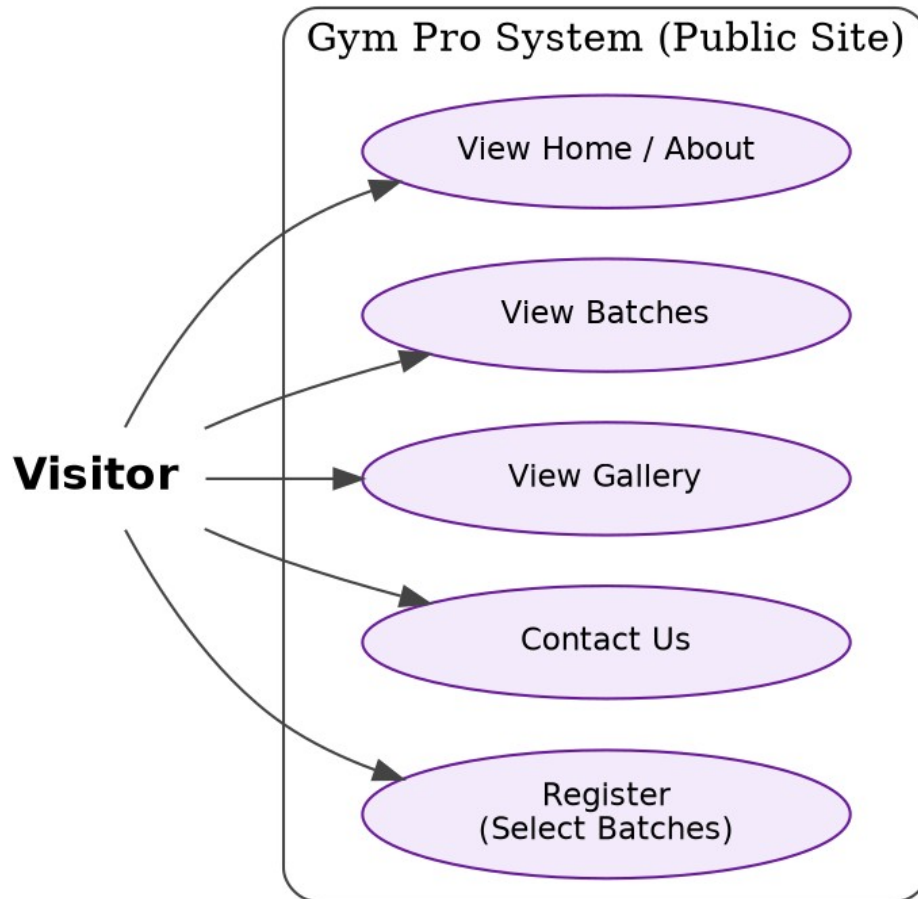


Fig 4.1(a): Use-case diagram – Visitor

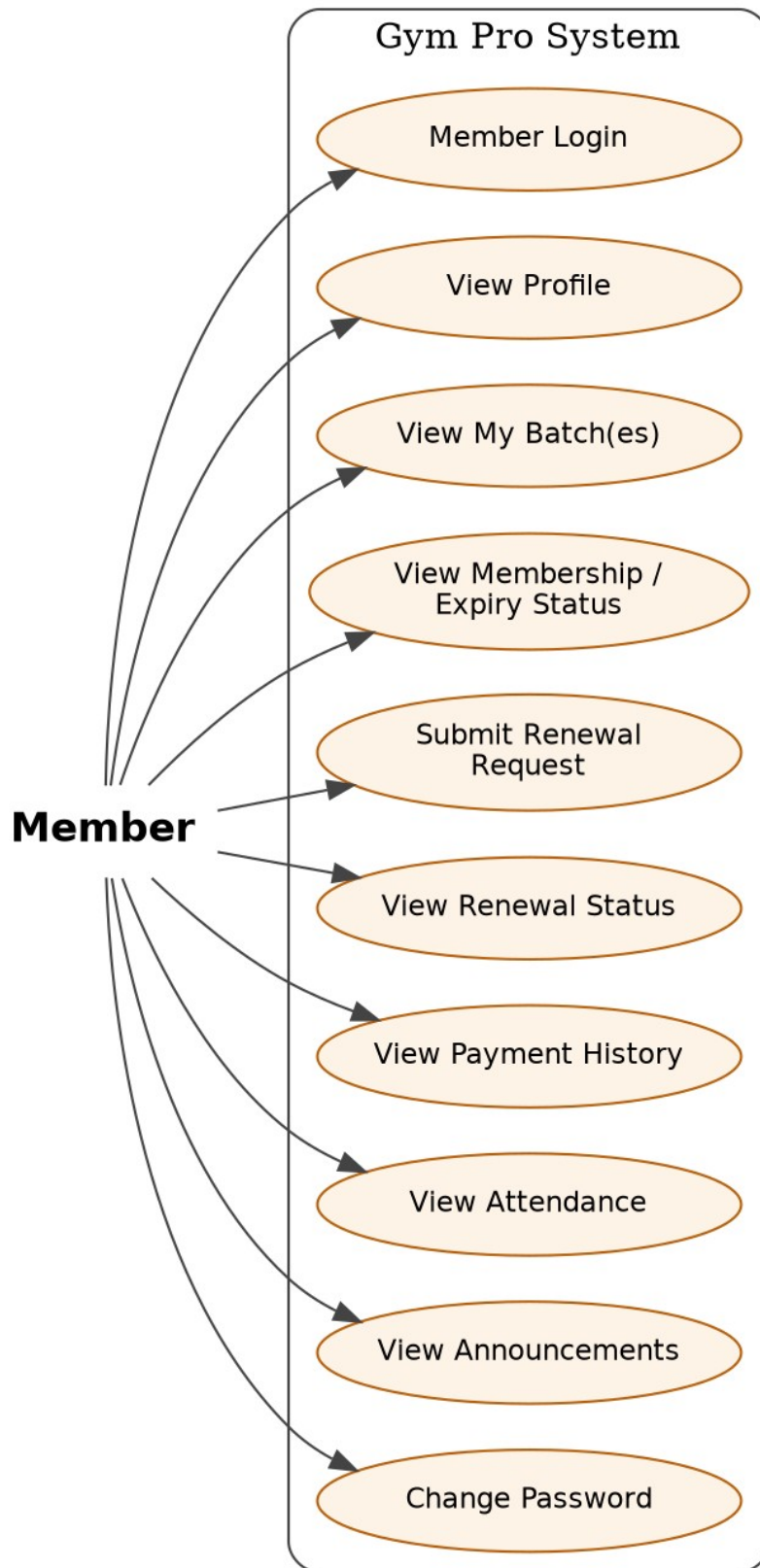


Fig 4.1(b): Use-case diagram – Member

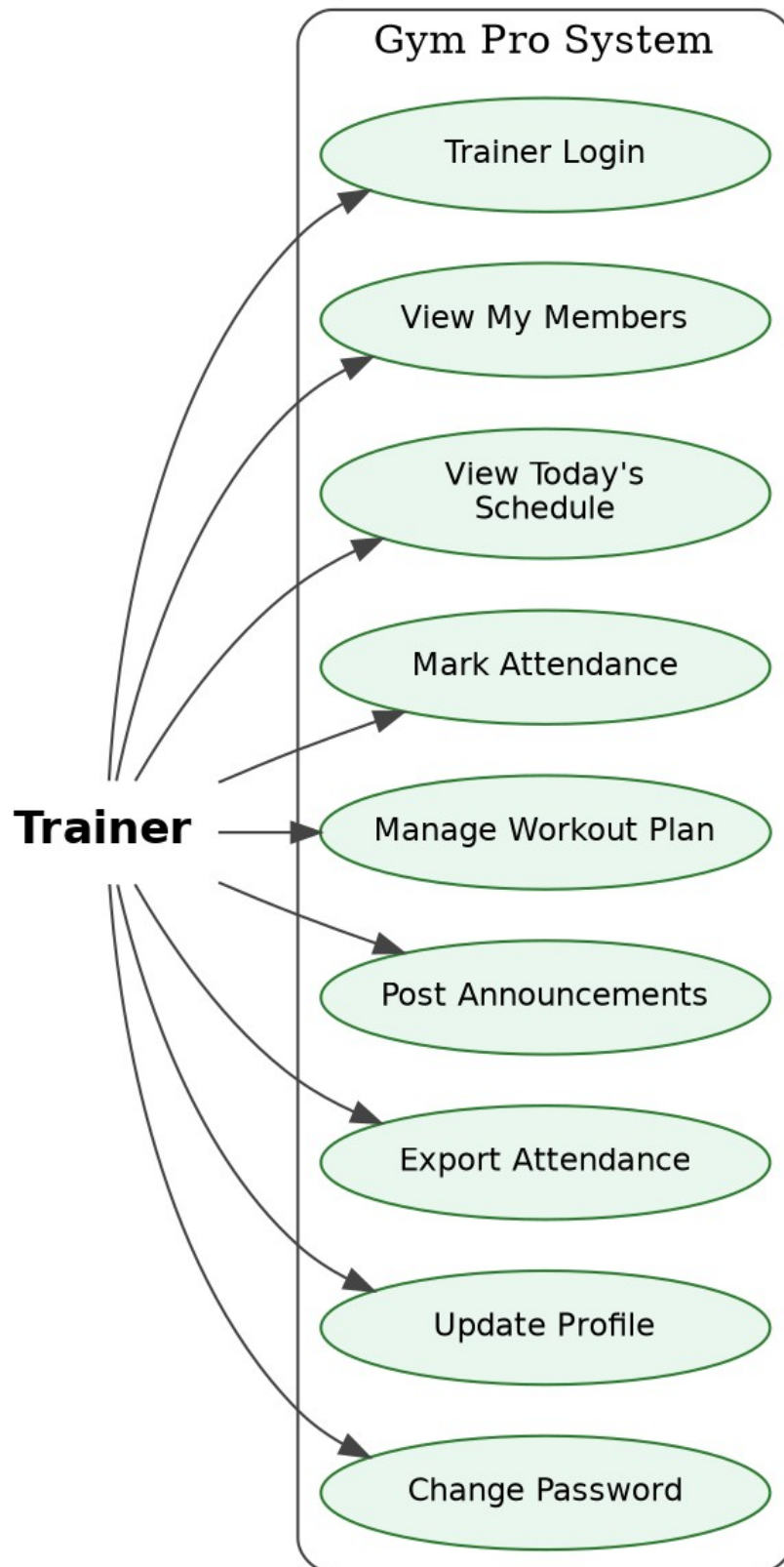


Fig 4.1(c): Use-case diagram – Trainer

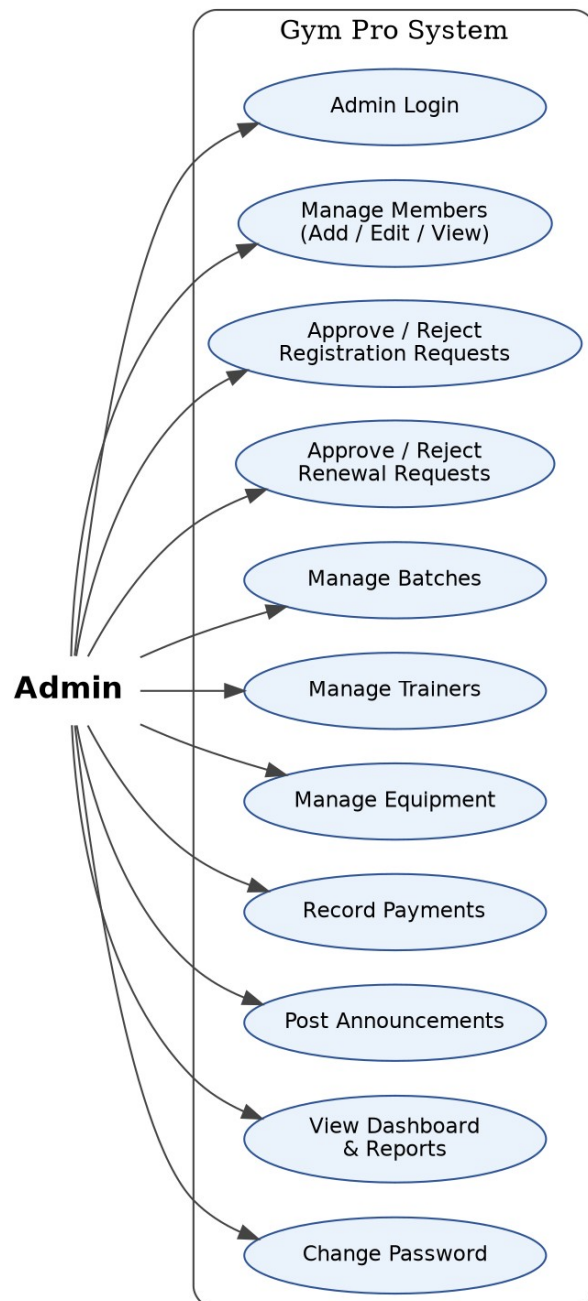


Fig 4.1(d): Use-case diagram – Admin

4.2 Data Flow Diagram

The Data Flow Diagram (DFD) shows how data moves between the external entities, the processes and the data stores of the system.

Level 0 – Context Diagram

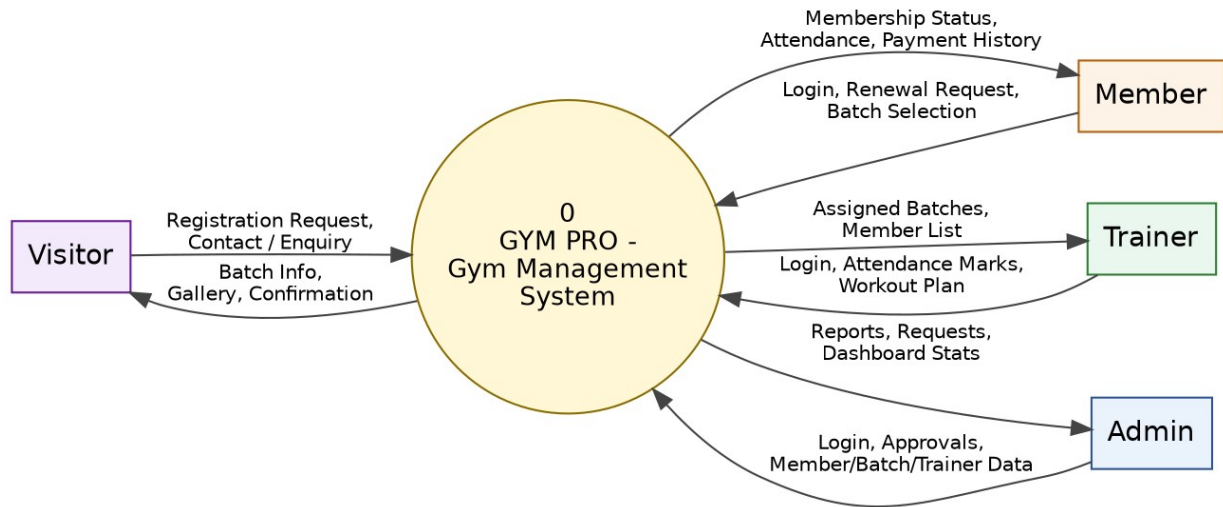


Fig 4.2(a): DFD Level 0 (Context Diagram)

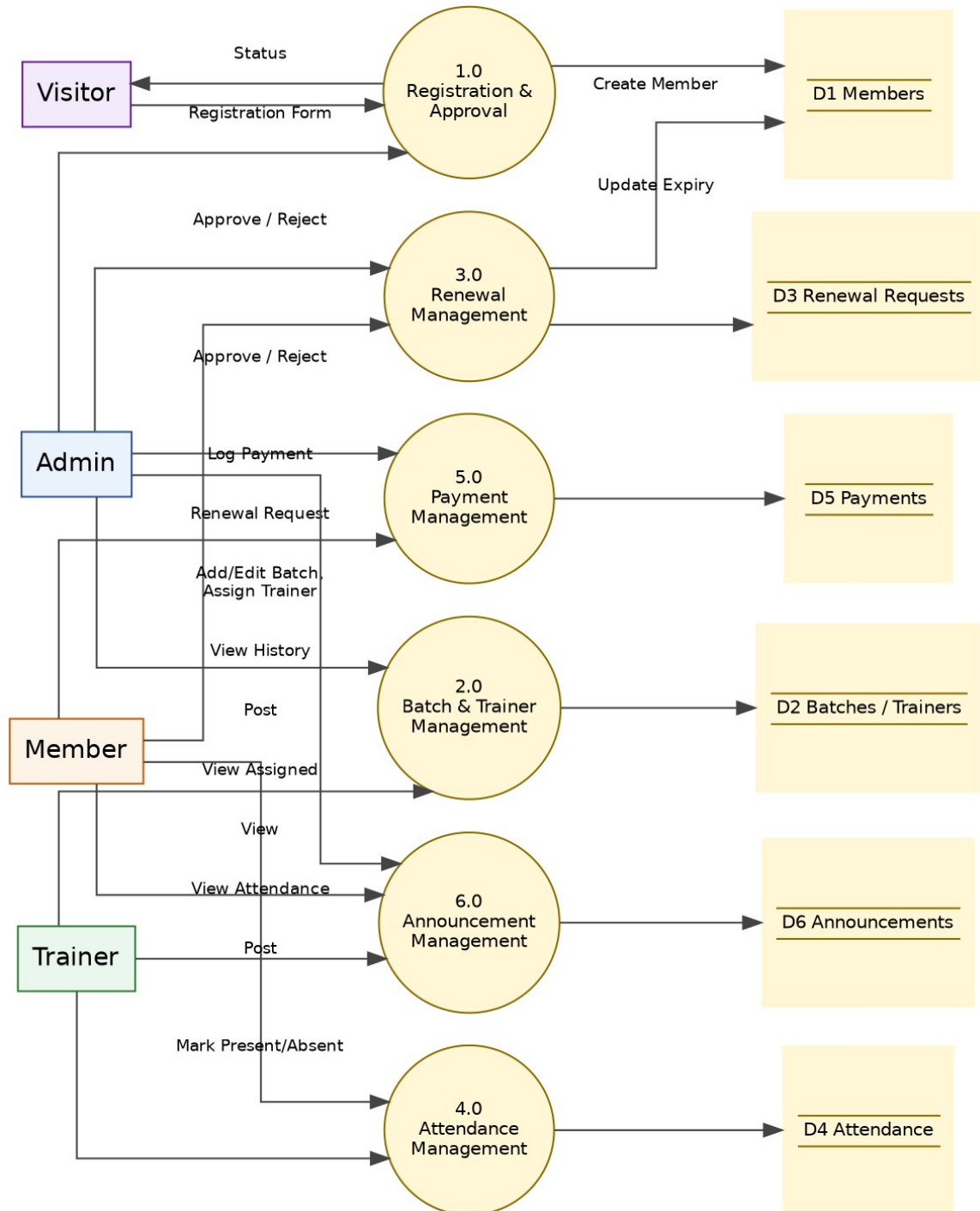
Level 1 DFD

Fig 4.2(b): DFD Level 1

4.3 E-R Diagram

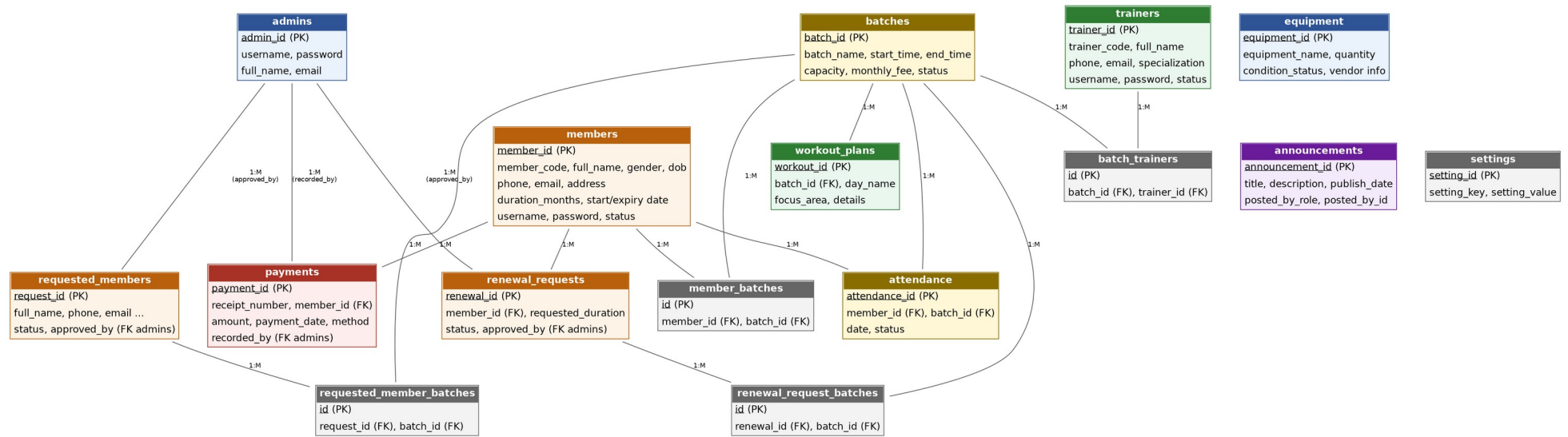


Fig 4.3: Entity-Relationship Diagram of GYM PRO database

4.4 Data Dictionary

The data dictionary below lists every table of the gym_pro database exactly as created in the system (16 tables in total), showing each column, its data type and its key/constraint.

admins

Column Name	Data Type	Key / Constraint
admin_id	INT (Auto Increment)	Primary Key
username	VARCHAR(50)	Not Null, Unique
password	VARCHAR(255)	Not Null (hashed)
full_name	VARCHAR(100)	Not Null
email	VARCHAR(100)	Not Null, Unique
created_at	DATETIME	Default Current Timestamp

trainers

Column Name	Data Type	Key / Constraint
trainer_id	INT (Auto Increment)	Primary Key
trainer_code	VARCHAR(20)	Not Null, Unique
full_name	VARCHAR(100)	Not Null
gender	ENUM('Male','Female','Other')	Default 'Male'
phone	VARCHAR(20)	Not Null, Unique
email	VARCHAR(100)	Not Null, Unique
specialization	VARCHAR(100)	-
photo	VARCHAR(255)	Default 'default-trainer.png'
username / password	VARCHAR(50) / VARCHAR(255)	Not Null, Unique / Not Null
status	ENUM('Active','Inactive')	Default 'Active'

batches

Column Name	Data Type	Key / Constraint
batch_id	INT (Auto Increment)	Primary Key
batch_name	VARCHAR(100)	Not Null
start_time / end_time	TIME	Not Null
capacity	INT	Default 20
monthly_fee	DECIMAL(10,2)	Default 0
status	ENUM('Active','Inactive')	Default 'Active'

batch_trainers

Column Name	Data Type	Key / Constraint
id	INT (Auto Increment)	Primary Key
batch_id	INT	Foreign Key -> batches
trainer_id	INT	Foreign Key -> trainers
(batch_id, trainer_id)	-	Unique Key

members

Column Name	Data Type	Key / Constraint
member_id	INT (Auto Increment)	Primary Key
member_code	VARCHAR(20)	Not Null, Unique
photo	VARCHAR(255)	Default 'default-member.png'
full_name	VARCHAR(100)	Not Null
gender	ENUM('Male','Female','Other')	Default 'Male'
dob	DATE	-
phone / email	VARCHAR(20) / VARCHAR(100)	Not Null, Unique (both)
address	VARCHAR(255)	-
emergency_contact	VARCHAR(20)	-
duration_months	INT	Default 1
membership_start_date	DATE	Not Null
membership_expiry_date	DATE	Not Null
username / password	VARCHAR(50) / VARCHAR(255)	Not Null, Unique / Not Null
status	ENUM('Active','Expired','Inactive')	Default 'Active'
created_by	VARCHAR(50)	-
created_method	ENUM('Request','Admin')	Default 'Admin'

member_batches

Column Name	Data Type	Key / Constraint
id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
batch_id	INT	Foreign Key -> batches
(member_id, batch_id)	-	Unique Key

payments

Column Name	Data Type	Key / Constraint
payment_id	INT (Auto Increment)	Primary Key

Column Name	Data Type	Key / Constraint
receipt_number	VARCHAR(20)	Not Null, Unique
member_id	INT	Foreign Key -> members
amount	DECIMAL(10,2)	Not Null
payment_date	DATE	Not Null
payment_method	ENUM('Cash','UPI','Card','Bank Transfer','Other')	Default 'Cash'
notes	VARCHAR(255)	-
recorded_by	INT	Foreign Key -> admins

requested_members

Column Name	Data Type	Key / Constraint
request_id	INT (Auto Increment)	Primary Key
full_name, gender, dob	VARCHAR/ENUM/DATE	full_name Not Null
phone, email	VARCHAR(20) / VARCHAR(100)	Not Null
address, emergency_contact	VARCHAR	-
duration	INT	Default 1
username / password	VARCHAR(50) / VARCHAR(255)	Not Null
status	ENUM('Pending','Approved','Rejected')	Default 'Pending'
request_date / approved_date	DATETIME	Default Current / Null
approved_by	INT	Foreign Key -> admins (nullable)

requested_member_batches

Column Name	Data Type	Key / Constraint
id	INT (Auto Increment)	Primary Key
request_id	INT	Foreign Key -> requested_members
batch_id	INT	Foreign Key -> batches

renewal_requests

Column Name	Data Type	Key / Constraint
renewal_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
requested_duration	INT	Default 1
status	ENUM('Pending','Approved','Rejected')	Default 'Pending'

Column Name	Data Type	Key / Constraint
	ejected')	
request_date / approved_date	DATETIME	Default Current / Null
approved_by	INT	Foreign Key -> admins (nullable)

renewal_request_batches

Column Name	Data Type	Key / Constraint
id	INT (Auto Increment)	Primary Key
renewal_id	INT	Foreign Key -> renewal_requests
batch_id	INT	Foreign Key -> batches

equipment

Column Name	Data Type	Key / Constraint
equipment_id	INT (Auto Increment)	Primary Key
equipment_name	VARCHAR(100)	Not Null
quantity	INT	Default 1
purchase_date	DATE	-
condition_status	ENUM('New','Good','Needs Repair','Old')	Default 'New'
vendor_name / vendor_location	VARCHAR	-
vendor_mobile	VARCHAR(20)	-

announcements

Column Name	Data Type	Key / Constraint
announcement_id	INT (Auto Increment)	Primary Key
title	VARCHAR(150)	Not Null
description	TEXT	Not Null
publish_date	DATETIME	Default Current Timestamp
posted_by_role	ENUM('Admin','Trainer')	Default 'Admin'
posted_by_id	INT	Not Null

workout_plans

Column Name	Data Type	Key / Constraint
workout_id	INT (Auto Increment)	Primary Key
batch_id	INT	Foreign Key -> batches
day_name	ENUM('Monday'...'Sunday')	Not Null

Column Name	Data Type	Key / Constraint
focus_area	VARCHAR(100)	-
details	TEXT	-
(batch_id, day_name)	-	Unique Key

attendance

Column Name	Data Type	Key / Constraint
attendance_id	INT (Auto Increment)	Primary Key
member_id	INT	Foreign Key -> members
batch_id	INT	Foreign Key -> batches
date	DATE	Not Null
status	ENUM('Present','Absent')	Default 'Present'
(member_id, batch_id, date)	-	Unique Key

settings

Column Name	Data Type	Key / Constraint
setting_id	INT (Auto Increment)	Primary Key
setting_key	VARCHAR(50)	Not Null, Unique
setting_value	VARCHAR(255)	Not Null

4.5 Flowchart (Optional)

The flowchart below traces the registration and approval process end to end, including the automatic overlap and batch-limit checks.

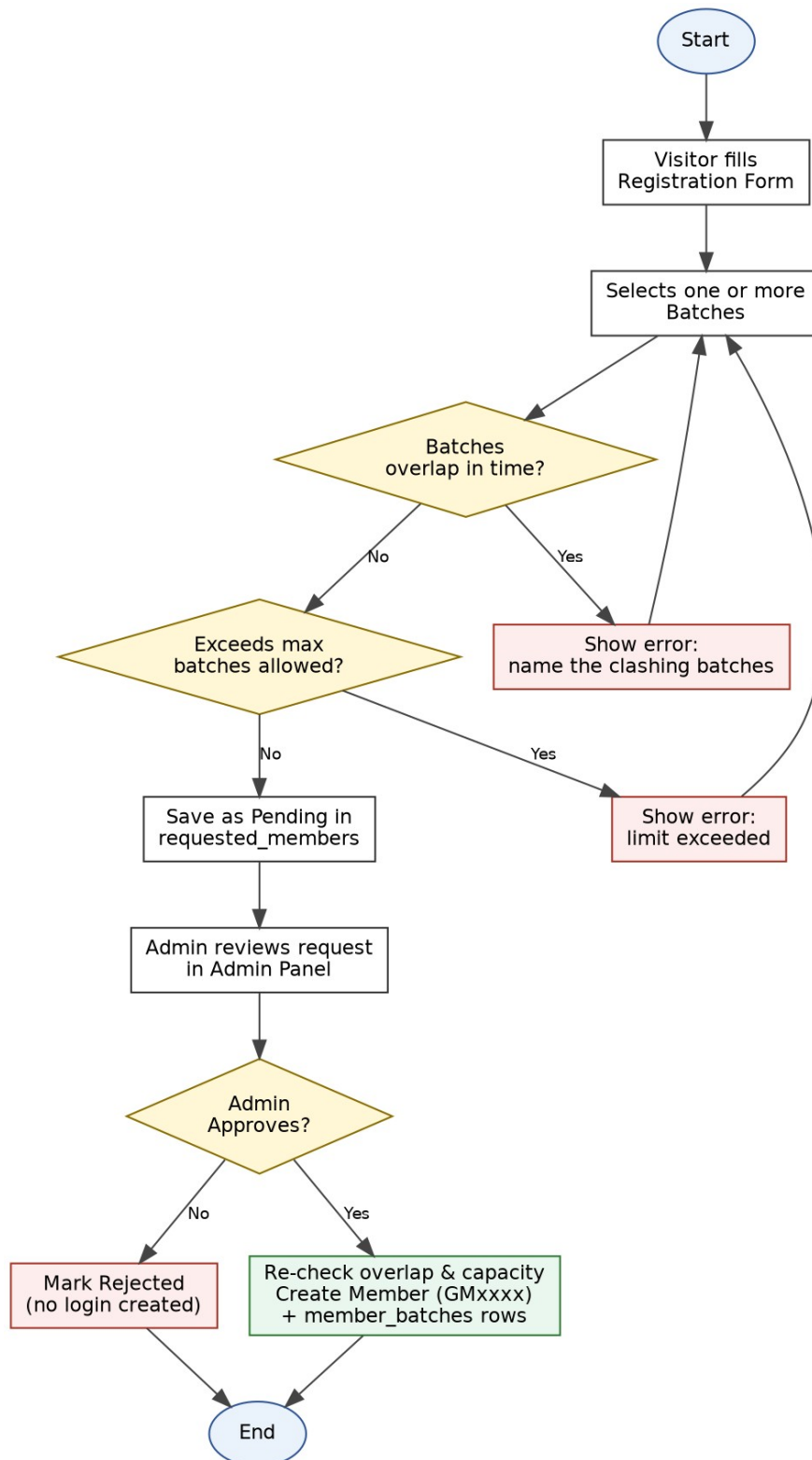


Fig 4.5: Flowchart of the Registration & Approval process

5. Screen Design

This section presents the layout (wireframe) of the important screens of GYM PRO – Gym Management System, covering the public website, the registration process and the three login panels.



Fig 5.1: Public Home Page

The public home page shows a welcome banner, quick highlights (expert trainers, multiple batches, modern equipment), a preview of active batches, and a footer with contact information. It is visible to every visitor without logging in.

GYM PRO	Member Login
----------------	--------------

Member Login

[eye icon]

New here? Register | Staff Login (settings.php)

Fig 5.2: Login Page

A single, simple login form is used for the Member panel; the Trainer and Admin panels use the same layout with their own username/password fields and route the user to the correct dashboard after checking the session role.

GYM PRO Public Registration	
Register - Personal Details	Select Batch(es) (grouped, overlap blocked) Morning ☒ Morning Yoga 06:00-07:00 ☐ Morning Strength 06:00-07:00 (conflict) ☐ Morning Cardio 07:15-08:15 Evening ☐ Evening Zumba 17:00-18:00 ☐ Evening Strength 18:15-19:15 Max 3 batches - 1/3 selected
Overlap error shown here if two selected batch timings clash Submit Registration Creates a Pending row - login is granted only after Admin approval	

Fig 5.3: Public Registration Page

The registration page collects the visitor's personal details on the left and lets them pick one or more batches on the right. Batches are grouped by time of day; selecting two batches whose timings overlap is blocked immediately and an error is shown, and the number of batches selected is checked against the maximum allowed.

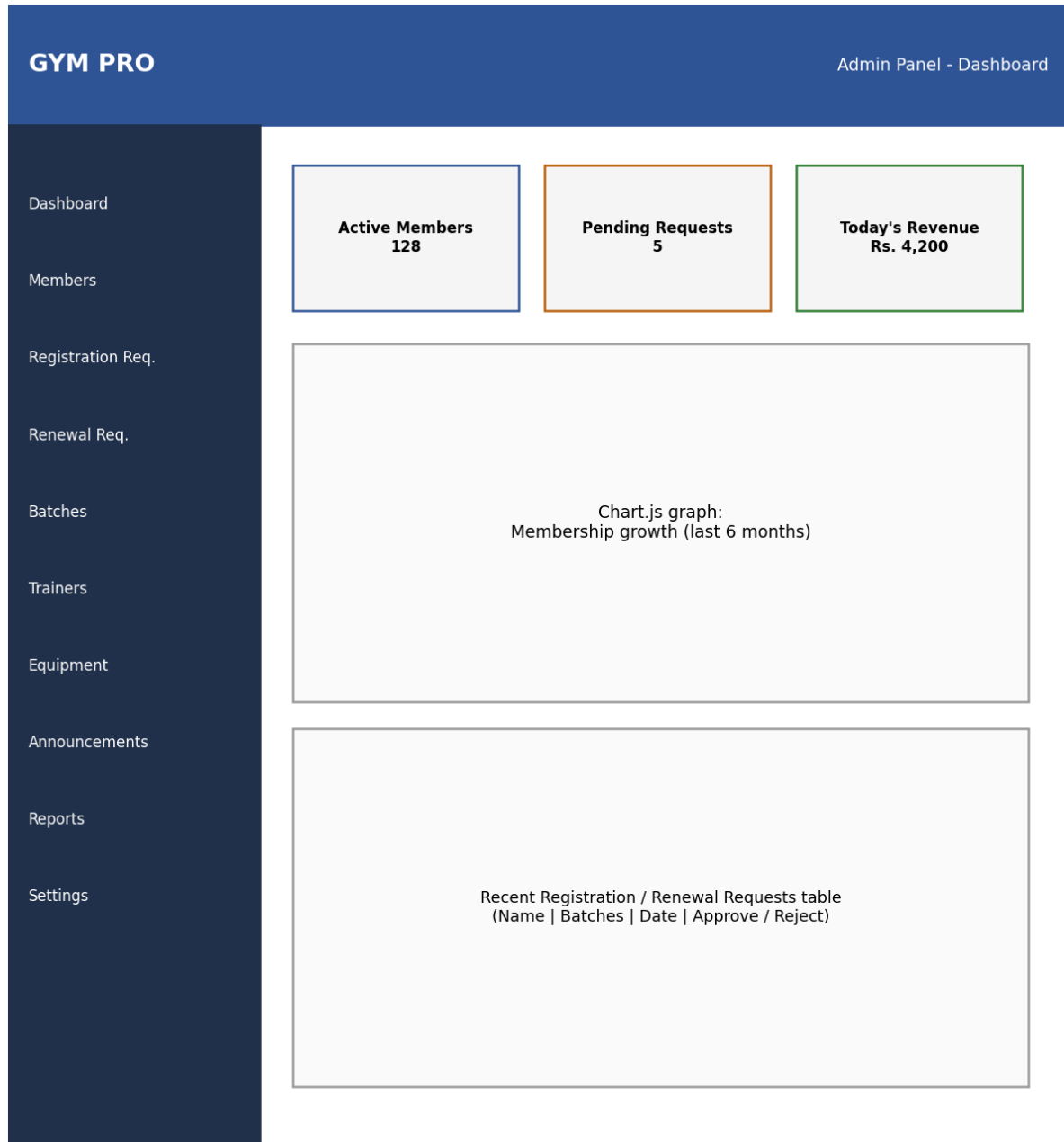


Fig 5.4: Admin Dashboard

The Admin dashboard shows quick-glance statistics (active members, pending requests, revenue), a membership-growth chart, and a table of recent registration/renewal requests that can be approved or rejected directly from the dashboard. A side menu gives access to every management module.

GYM PRO Member Panel

Dashboard My Batch Attendance Membership Payments Renew Announce Profile

Welcome, Sneha Patel (GM0002) | Status: Active | Expiry: 3 months left

My Batches

Morning Cardio | 07:15 - 08:15 | Trainer: Rahul Sharma | Rs.700/mo

Evening Zumba | 17:00 - 18:00 | Trainer: Priya Nair | Rs.900/mo

Recent Attendance

Date	Batch	Status
(Present / Absent list, last 7 days)		

Announcements

Welcome to Gym-Pro - join more than one batch if timings do not overlap.

Fig 5.5: Member Dashboard – My Batch

A member sees their own status and expiry date, the list of batches they are enrolled in with the assigned trainer and fee, their recent attendance, and any announcements posted by the Admin or trainer.

GYM PRO

Trainer Panel - Attendance

Dashboard

My Members

Today's Schedule

Attendance

Workout Plan

Announcements

Profile

Batch: Morning Yoga (v)

Date: 12-09-2026

No	Member	Code	Status
1	Amit Verma	GM0001	(o) Present () Absent
2	Sneha Patel	GM0002	(o) Present () Absent

Save Attendance

Export Attendance (CSV) available via [export-attendance.php](#)

Fig 5.6: Trainer – Mark Attendance

A trainer selects one of their assigned batches and a date, marks every member Present or Absent, and saves the attendance in one action. Attendance can also be exported for record-keeping.

6. Testing (Manual, Test Cases and Test Data)

6.1 Manual Testing

Before the system was put into use, every module was tested manually by entering both valid and invalid data and observing whether the result matched what was expected. The following modules were tested in this way:

- Registration form — valid data, duplicate phone/email, and overlapping/over-limit batch selection.
- Login for all three panels (Admin, Trainer, Member) — correct credentials, wrong password, wrong username, and blank fields.
- Batch overlap validation — pairs of batches with overlapping, back-to-back and non-overlapping timings.
- Renewal request and approval — valid renewal, renewal against a full batch, and renewal for an already-expired member.
- Attendance marking, payment recording and announcement posting — checked for correct saving and correct display back to the member/trainer.

Each test was repeated with more than one set of input values (see the test cases below), and the actual result was compared against the expected result before the module was considered complete.

6.2 Test Cases

Test Case 1 – Member Registration

Field	Input 1	Input 2	Input 3	Input 4
Full Name / Phone / Email	Valid, all new	Phone already used	Valid, all new	Valid, all new
Username	Unique	Unique	Unique	Unique
Batch(es) selected	Non-overlapping, within limit	Non-overlapping, within limit	Two batches overlap in time	Batches selected exceed limit
Result	Pass – Pending request created	Fail – "already registered" error	Fail – overlap error shown, blocked	Fail – "limit exceeded" error

Test Case 2 – Login (Admin / Trainer / Member)

Field	Input 1	Input 2	Input 3	Input 4
Username	Correct	Correct	Incorrect	Blank
Password	Correct	Incorrect	Correct	Blank
Result	Pass – redirected to dashboard	Fail – "invalid credentials"	Fail – "invalid credentials"	Fail – "required field" validation

Test Case 3 – Batch Overlap Validation

Field	Input 1	Input 2	Input 3	Input 4
Batch A timing	06:00 – 07:00	06:00 – 07:00	06:00 – 07:00	06:00 – 07:00

Field	Input 1	Input 2	Input 3	Input 4
Batch B timing	17:00 – 18:00	06:00 – 07:00	06:30 – 07:30	07:00 – 08:00
Overlap?	No	Yes (identical)	Yes (partial)	No (back-to-back)
Result	Pass – both allowed	Fail – selection blocked	Fail – selection blocked	Pass – both allowed

Test Case 4 – Renewal Request Approval

Field	Input 1	Input 2	Input 3	Input 4
Member status	Active, expiring soon	Active, expiring soon	Active, expiring soon	Already Expired
Requested batch(es)	Same, has room	Now at full capacity	Overlaps another selection	Same, has room
Result	Pass – Approved, expiry extended	Fail – Rejected (capacity)	Fail – Rejected (overlap re-checked)	Pass – Approved, status back to Active

7. Enhancements

7.1 Advantages

- All database queries use prepared statements (mysqli), which protects the system against SQL-injection attacks.
- Passwords are stored using PHP's password_hash()/password_verify() functions and are never kept in plain text.
- Each panel — Admin, Trainer and Member — has its own independent session-based login; an unauthorized visitor is redirected back to that panel's login page.
- The core business rule — that a member cannot join two batches whose timings overlap — is enforced on the server side, so it cannot be bypassed even if JavaScript is disabled in the browser.
- A trainer can view and manage only the batches actually assigned to them, not the entire gym's data.
- The system is built to run on both modern PHP (7.x / 8.x) and older PHP 5.4 installations, which is useful in college lab environments.
- A simple, responsive Bootstrap 5 interface works acceptably on both desktop and mobile screens.

7.2 Limitation

- There is no online payment gateway; every payment is collected in person and simply logged into the system by the Admin afterwards.
- Photo upload for members and trainers is not yet implemented, although the database already has a column reserved for it.
- No automatic email or SMS notification is sent when a registration or renewal request is approved or rejected — the member has to log in to check the status.
- Long lists (members, trainers, requests) do not yet have pagination, which may slow down as the number of records grows very large.
- The Contact Us form on the public site only shows a confirmation message; enquiries are not yet stored in the database for the Admin to review later.

7.3 Future Scope

- Add real photo upload for members and trainers using PHP's move_uploaded_file() function.
- Integrate an online payment gateway (for example Razorpay or a UPI deep-link) so that members can pay their fee directly from their dashboard.
- Send an automatic email or SMS notification when a registration or renewal request is approved or rejected.
- Add a contact_messages table and an Admin inbox so enquiries from the Contact Us page can be tracked and answered.
- Add pagination and search/filter options to all the long list pages.

- Package the member-facing pages as a Progressive Web App (PWA) so members can check their schedule and attendance conveniently on mobile.

8. Bibliography

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<https://www.php.net>

<https://www.w3schools.com>

<https://getbootstrap.com>

<https://dev.mysql.com/doc/>

<https://www.chartjs.org>

Books:

- PHP and MySQL Web Development – Luke Welling & Laura Thomson
- Learning PHP, MySQL & JavaScript – Robin Nixon